

EE1060: Differential Equations and Transform Techniques - Quiz 1

Name: _____

Roll No: _____

Instructions

1. The quiz duration is 30 minutes, starting at 11:15. Ensure that you return the paper by 11:45. Please be aware that **late submissions** may incur **penalties**.
2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.

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1. (2 points) Determine if the differential equation $\cos x \frac{d^2y}{dx^2} + \frac{dy}{dx} - xy = 0$ is linear and identify its order, providing supporting arguments for linearity or otherwise.

2. (2 points) Determine an explicit parametric solution for the ODE $\frac{dx}{dt} = -\frac{k}{x^2}$, where k is a constant.

3. (2 points) Determine the values of α and β for which $y = ce^{-bx} + \alpha x + \beta$ (where c is an arbitrary constant) is a solution to the ODE $\frac{dy}{dx} = ax - by$.

4. (2 points) Determine the maximum value of r for which $y = x^r e^{\alpha x}$ is a solution to the differential equation $\frac{d^2 y}{dx^2} + 2\alpha \frac{dy}{dx} + \alpha^2 y = 0$.

5. (2 points) Determine a one-parameter solution to the ODE $\frac{dx}{dt} + kx^2 \sin(bt) = 0$, where k and b are constants.